SPACE RESEARCH AND EXTRATERRESTRIAL CONSTRUCTION INDUSTRY

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***Abstract*— Advance of civilization has been realized through the ages by exploration of unknown parts of the Earth and by exploiting new resources thus acquired. There are currently practically no new areas on this planet to be explored and exploited. This is achieved by the completion of the period called globalization. As a natural continuation of this process, humans will now explore and exploit new areas beyond Earth, i.e. in space. Moon has already been explored partly, it awaits a complete exploration and colonization. The next step will be Mars and beyond. Habitation on these new space bodies will firstly necessitate construction of outposts on these bodies. Thus construction on the Moon, Mars, and other space bodies can be considered to be an important component of space research. In the present article, an overview of space research is presented together with the innovations awaited in space construction industry and the importance of feedback of these studies to construction industry in general.**

Keywords—Space, Earth, Moon, Mars Civil Engineering.

I. INTRODUCTION

First environment of humans was a few kilometers around their habitats. They were exploiting the resources in that environment. They knew the characteristics of the "nature" which was prevailing in that area.

As human groups, clans widened their environmental area, the resources became richer and they met new "natures". Richer resources enabled them to acquire a higher level of economy; meeting new natures made them more flexible and more powerful to deal with new difficulties thus increasing their technological abilities. Reaching higher levels of economy and technology gave them the power for conquering new areas thus widening their "world". This process, working through ages in the form of a continuous loop (see Figure 1), can be thought to be led to what is now called as "globalization" that may be considered as completed in 2000's. Men, starting from small portions of the Earth, now have conquered almost totality of the planet on which they are living, they are accustomed to all "natures" prevailing, exploiting nearly all resources on it. Their world now is the whole globe and they are now conscious that they are living on a globe in 3 dimensions. The loop shown in Fig. 1 will now lead humans to extraterrestrial places.

II. SPATIALIZATION - WORLD 2050

The Russian scientist Konstantin E. Tsiolkovsky (1857-1935) had said "The Earth is the Cradle of Mankind, but one cannot expect to remain forever in the cradle". His epitaph, written by himself reads: "Man will not always stay on Earth; the pursuit of light and space will lead him to penetrate the bounds of the atmosphere, timidly at first, but in the end to conquer the whole of solar space". Indeed "man goes everywhere he can go". As explained above through the unending loop that brought mankind into globalization, the capacity of men for going to new lands increases endlessly as new worlds are added to existing one, which results in going to new lands. Thus, going to space and conquering new additions to the existing lands is nothing but a natural continuation of what man made during all time passed, following the saying "Man goes everywhere he can go".

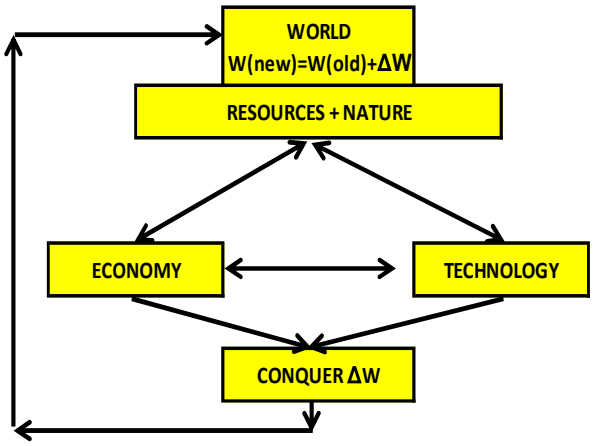

Fig. 1. Effect of enlargening "world" of humans on advance of economy and technology

Man's conquest can be said to be multi-directional. If one considers the conquest until recent times, it is on the surface of the world, it is towards the center of the Earth at land and at seas, and on altitude, away from the surface of Earth. In Figure 3 is shown the history of man's conquest in altitude. Just after

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1900's, with invention of balloons and planes, many altitude records had been broken until 1961, reaching 51700 meters. That distance was surpassed (reaching 315000 m) by the first space flight that Soviet cosmonaut Yuri Gagarin had realized the same year. That record had been broken several times until 1968 where men have gone up to 1369 km in his flights around the Earth. And then came the time of making tours around the Moon and landing on it, which brought the record to 400171 km from Earth in 1970. All these events can be seen in Fig 3, with a continuous line. This record has not been broken yet, and it seems that it will not be broken until 2050's that will bring humans to more than 200 million of km's "above" Earth. This jump will correspond to going to Mars [1].

Fig. 2. The current "world" of humans. (The picture of the Earth and Moon in a single frame, the first of its kind ever taken by a spacecraft, recorded at September 18, 1977, by NASA's Voyager 1 when it was 12 million km from Earth. < http://grin.hq.nasa.gov/ >)

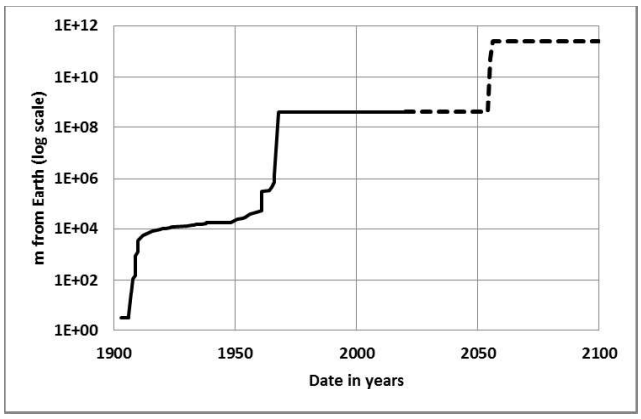

Fig. 3.Manned flight records from Earth

Thus, one can say that the world of humans in the years 2000 is the Earth, the Moon, and the space between Moon and Earth, although Moon is still waiting to be explored more, inhabited, and exploited. The world of mankind in the years 2100 will consist of Earth, Moon, Mars, probably some of the satellites of Mars, and the space between Earth and Mars (Figure 4). It is highly probable that the years after 2200's will add to this world Jupiter, some of the satellites of Jupiter and the space between Jupiter and Mars.

III. Pioneers of Going to Moon and Mars

Currently there are more than 70 countries and unions who have a "space program" of some sort, and only 10 of them have proved their capacity of launching satellites into space: US, Russia, Ukraine, EU, China, India, Iran, Israel, Japan, and North Korea. When it comes to sending humans to space, that number goes down to three, counting only US, Russia and China. It is also worthy to note that only US was capable of sending humans to a space body, to our moon namely. It is expected that the number of countries with manned missions to the Moon will have increased until 2030, and there will be at least one manned mission to Mars around 2050. It is considered that the two recent political events, namely Brexit, exit of United Kingdom from European Union, and the election of Mr. Donald Trump as new US president, will not affect too much the expectancies given above.

Thus, it is very probable that in the coming 4 decades humans will have some outposts on the moon and even on Mars enlarging their "world" (Figure 4). Previous visits to moon were for short periods, but these new ones will certainly be for long periods, if not as permanent settlements. The moon is visited several times years ago, and there is no meaning of making new short trips to it, except for new-comers. Mars is so far away that it is not meaningful to go there for just a short time, considering also the time windows making that trip feasible.

It is worth to mention here that besides Moon and Mars, there are plans also to realize manned missions to Phobos, a satellite of Mars [1]. Another interesting point is that currently there are now private organizations that are planning to launch rockets to realize manned missions to Mars (https://en.wikipedia.org/wiki/List_of_private_spaceflight_companies).

Fig. 4. "World" of humans in next future: Earth, Moon, Mars and the space between them

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IV. Humans on Space Bodies

A very important aspect of space bodies, at least of those that humans can see and observe currently, is their hostile environment. Indeed, on none of the space bodies mentioned in this article, men cannot live even a part of a second without taking special measures, due to differences between the atmosphere of Earth and atmosphere of these bodies. That means that on those bodies a different nature is prevailing. On the other hand, the resources on them are different from those that are on Earth. Thus, going there, living under their local nature, and exploiting their resources will have an immense positive effect on our capabilities and flexibility, and also will put our economies at a much higher level. The feedback of that space research and applications on those space bodies will make humanity richer and stronger. Comparisons with new areas conquested and Earth will help humanity to understand better the old areas inhabited as explained in the phrase "We went to explore the Moon, and in fact discovered the Earth" by Eugene Cernan, Apollo 17 astronaut.

It seems that, until now, there have been more than 30000 innovations coming from man's attack towards skies beyond atmosphere. Some of them are directly related to space applications, but most of them are indirectly related to space research. Many of them have helped to give birth to unexpected by-products which can be used in areas not related to space affairs, thus benefiting also humans who are not related to space affairs in no way.

This means that space research is not a single dimensional activity. On the contrary, it is a multi-dimensional task if not infinite dimensional. This is true for the aims and for outputs at the same time. The feedback of space research is very important for all mankind.

V. Habitation / Construction On Space Bodies

As it is said in above paragraphs, space research is a multi-dimensional task. One dimension of it is founding habitations, i.e., bases on Moon, Mars, and other extraterrestrial bodies. Construction is an important item in founding these outposts. Obviously, many innovations will be done in performing construction operations on these bodies, which will certainly benefit humans' terrestrial life.

Construction on space bodies will be carried on by the next generation of civil engineers, of course in close collaboration with engineers and scientists from many other disciplines [2,3]. They will act according to the definition of civil engineering which is given by Thomas Tredgold back in 1828: "That species of knowledge which constitutes the profession of Civil Engineering; being the art of directing great sources of power in Nature for the use and convenience of man, …". The main idea in this definition is directing natural resources found on these space bodies for the use of man in order to build structures of all kinds [4].

In inhabiting space bodies, the first habitations will certainly be those brought from Earth. Carrying 1 kg of material from Earth to Moon for instance, costs about 20000 US$. That price is much higher for Mars and beyond. Even if this rate could be reduced in the future, it will still be high and always risky. So, all possible ways of making maximum usage of in-situ materials will be tried on space bodies [5]. In fact, there will be 3 ways of using in-situ materials:

- Use raw in-situ material as much as possible.
- Find effective and simple ways of treating in-situ material for being used in constructions.
- Develop techniques for advanced treatment of in-situ materials for extracting elements and minerals.

Among these, the third alternative of course cannot be realized at early times, but the first two can take place at very early stages.

A. *In-situ materials and plants on space bodies*

In fact 5 types of products can be considered to be used on space bodies, as shown in Table 1. The 1st type is a product completely produced on Earth, with terrestrial materials. 2nd types of materials are those produced on space bodies using terrestrial resources. 3rd types are those produced at plants on space bodies using resources coming from both Earth and space body. 4th types are produced on space body using in-situ resources. The fifth ones are partly produced on Earth and on space body, using resources from Earth and space body. The evolution of habitats produced in these 5 manners will somehow be as shown in Figure 5. This figure shows that the proportion of products 1 and 2, which are produced using resources coming from only Earth will be reduced as time goes on, and the proportion of those which are produced on space bodies will increase with time.

An estimation of the quantities of terrestrial materials transported from Earth to space body, and in-situ materials, shown in Figure 6 shows that as time goes on, quantity of materials obtained from in-situ resources will increase while the inverse will take place for terrestrial materials.

TABLE I. Types Of Products On Space Bodies

PRODUCT	RESOURCES		PLANT	
	EARTH	S. BODY	EARTH	S. BODY
1	X		X	
2	X			X
3	X	X		X
4		X		X
5	X	X	X	X

It is worth to note here that the first uses of in-situ materials on space bodies will be the coverage of structures by regolith as a protection from radiation, construction of walls and structures by reinforced regolith, and forming structures by gabions filled with in-situ dust.

B. *A better understanding of structures*

On Earth the primordial loads are gravitational. Environmental live loads (earthquakes, wind, snow ...) may also be important. On space bodies in consideration the gravitational loads are as important as on Earth due to shielding materials. But effects coming from inside pressure is much

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more important. Cracks are not allowed. These changes will lead to new innovations in construction industry. Other sources of changes will be [6-9].

- Constructions in lava tubes
- Habitats embedded in regolith
- Tensegric structures
- Structures with reinforced regolith walls
- Inflatable structures
- Extensive use of prestressing
- Structures with no stress concentrations

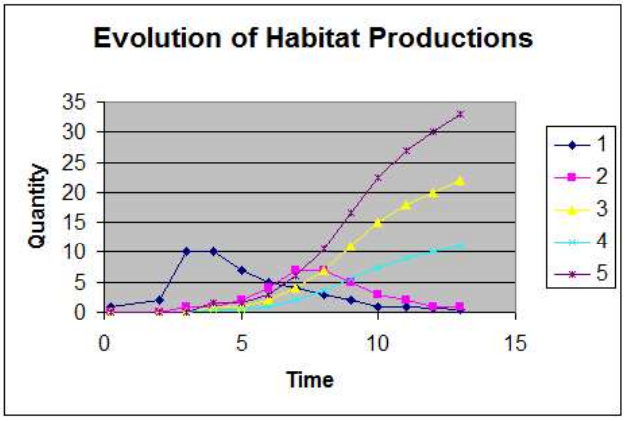

Fig. 5. Evolution of habitat productions on space bodies

Because of the hostile environment on space bodies, the robustness of structures constructed on Moon and elsewhere will be very important. They will be designed for all types of uncertainties and unattended effects like meteorite impacts up to a certain size. Structural smartness will be an important measure against uncertainties [10]. The scheduling of their construction, the process of construction, and their maintenance must all follow the principles of robustness [5, 11-13].

C. *Automated construction*

Due to the difficulties and dangers in using manual workmanship, automated mining, earthwork and construction techniques will be advanced and used on extraterrestrial bodies. Many kinds of robots will be developed for special works. Construction industry will be fully automated.

D. *Special construction machines*

Special environmental lunar and Martian conditions will enforce development of special construction machines. They will be electrically operated, fully automatic and/or tele commanded, lighter, and with special measures against dust. Their lightness will certainly reduce their effectiveness; therefore special measures will have to make them powerful despite this deficiency.

E. *Nonlinearity*

On Earth, construction materials are used in linear zone for being on the safe side and also due to the computational difficulties coming from nonlinearity. On space bodies, the designers will not have that luxury because of very high material costs and difficulties in supply. All materials will be exploited up to their maximum limits, of course without concession from safety [14].

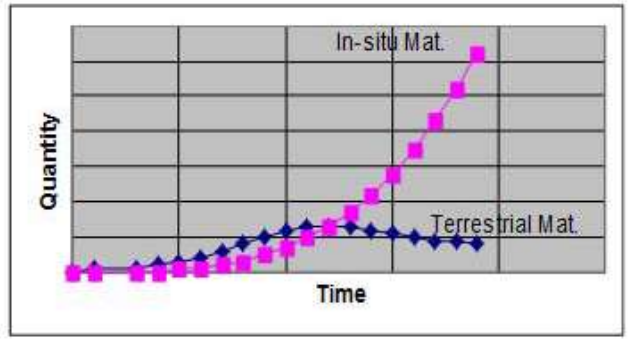

Fig. 6. Estimated use of in-situ materials and terrestrial materials on space bodies.

F. *Building Information Systems*

All the structures will be controlled, managed by a central management system. There will be sensors measuring pressure, temperature, humidity, light, oxygen content, CO2, humidity, stresses and strains in the structures, etc. at every possible zone of the structures. All this data will be saved and evaluated in the managing systems. The managing system will decide on actions to be taken according to this data, taking into account the most effective use of resources (energy and materials), comfort, and security in the base. According to the decisions taken by the central management system, through actuators, the necessary actions will be taken. This means that all structures will be smart, or intelligent. In fact they will be acting as robots, or in other words, as mechatronic systems. The experience gained through these all these applications will certainly benefit terrestrial applications.

G. *Waste treatment*

There will be a very advanced waste management system. All liquid, solid and gaseous wastes will be treated to a maximum extent and recycled for repeated use.

CONCLUSIONS

It is to be noted that, up to now, the pioneering countries have been extremely successful in their manned an unmanned operations on the moon and elsewhere in the space. It is this success that gives all mankind the hope and the security that the coming operations will also be successful and thus the thought that this pioneering work has to be followed. In this context, civil engineers will continue to do their work adopted for the new Nature and carry back their innovations to Earth.

The loop mentioned in above paragraphs, pointing to reaching better levels of living through meeting new environments and new resources has always been an adventure. The one in front of humanity is evidently the greatest of them. It is in fact a natural continuation of the historical activities starting from primitive habitats to globalization and it marks the passage from globalization to spatialization, or from planet Earth to space bodies. The preceding steps were taken by hazard or by partial plans. In contrast to them, the next step

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will have to be taken very carefully, with fully elaborated plans. Construction is only a small part of it. The success of that step will completely change the life of humans due to feedback coming from innovations which have to be realized in order to be successful in these hostile and far environments.

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